

Clarkson Institute Summer School: Examples Sheet 3 - Subgroup theory

Please send all comments and corrections to jmm232@cam.ac.uk.

Definition of a subgroup

1. Find all the subgroups of D_4 and describe them geometrically. Are the subgroups of order 4 all isomorphic to one another?

Solution: Let $D_4 = \{e, r, r^2, r^3, m, rm, r^2m, r^3m\}$. From smallest to largest subgroups, we have:

- The subgroup of order one is $\{e\}$. This is just the identity transformation.
- The subgroups of order two are the reflection subgroups, $\{e, m\}$, $\{e, rm\}$, $\{e, r^2m\}$ and $\{e, r^3m\}$. These all involve breaking all the symmetry of the square, except for reflectional symmetry in one of the axes (either horizontal, vertical, or one of the two diagonals). In addition, there is also the subgroup $\{e, r^2\}$ consisting of a rotation by π ; this involves choosing an orientation for the square (e.g. putting arrows on the sides) and also breaking the $\pi/2$ rotational symmetry - e.g. by colouring the edges alternating red and blue.
- There are three subgroups of order four. The rotations $\{e, r, r^2, r^3\}$ form a subgroup, corresponding to adding an orientation to the square (e.g. by adding arrows to all the sides). There is also a subgroup of reflections and rotations $\{e, m, r^2m, r^2\}$, which consist of reflections in the horizontal and vertical, together with a rotation by π . This implies breaking of symmetry in one direction, e.g. extending the square into a rectangle in one direction. Similarly there is a subgroup of reflections and rotations $\{e, rm, r^3m, r^2\}$, which consists of reflections in the diagonals, together with a rotation by π . This again implies breaking of symmetry in one direction; this is the symmetry group of e.g. a kite (stretching the square in one direction).
- Finally, there is the improper subgroup D_4 of D_4 .

The subgroups of order four are not all isomorphic. The rotation subgroup is isomorphic to C_4 , meanwhile the two subgroups involving reflections are both isomorphic to D_2 .

2. Let G be a group. Find an example of a subset $H \subseteq G$ which can be made into a group, but is not a subgroup of G .

Solution: Take $G = \mathbb{R}$ and take $H = \mathbb{R}^*$. Then $\mathbb{R}^* \subseteq \mathbb{R}$ is a subset of $\mathbb{R}$, and both a groups with respect to addition or multiplication as appropriate. However, $\mathbb{R}^*$ is not a subgroup of $\mathbb{R}$ because $\mathbb{R}^*$ does not contain the additive identity 0 of $\mathbb{R}$, so it cannot be made into a group with respect to the same operation as $\mathbb{R}$.

3. Prove that the set of elements of finite order in an Abelian group is a subgroup. Must this be the case in a non-Abelian group?

Solution: Let G be an Abelian group, and let H be the set of elements of finite order. We check each of the group axioms in turn:

(G1) *Closure.* Let $x, y \in H$ have finite orders n, m respectively. Then:

$$(xy)^{nm} = x^{nm}y^{nm} = (x^n)^m(y^m)^n = e,$$

using the fact that the group is Abelian in the second step. Thus xy has finite order, so H is closed under the group multiplication.

(G2) *Associativity.* This is naturally inherited from the larger group; since G is associative, any subset's multiplication is associative.

(G3) *Identity.* The identity has finite order, since it has order one, so is included in H .

(G4) *Inverses.* We saw on Examples Sheet 2 that an element and its inverse have the same order. Hence if $x \in H$ has finite order, we have that x^{-1} has finite order so is also in H .

Hence, indeed H is a subgroup.

This need not be the case in a non-Abelian group, because the product of two elements of finite order may have infinite order (i.e. the subset may not be closed). In order to find an example, we need an infinite group. The simplest example comes from matrix groups:

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1/2 \\ 2 & 0 \end{pmatrix}.$$

We observe that A, B both have finite order since:

$$A^2 = I, \quad B^2 = I.$$

But, the product AB has infinite order, since:

$$AB = \begin{pmatrix} 1/2 & 0 \\ 0 & 2 \end{pmatrix}.$$

The powers of this matrix are:

$$(AB)^k = \begin{pmatrix} 1/2^k & 0 \\ 0 & 2^k \end{pmatrix}.$$

4. Suppose that a group G has subgroups $H, K \leq G$. Is it true that $H \cap K$ and $H \cup K$ are subgroups? Give a proof or counterexample as appropriate.

Solution: It is certainly true that $H \cap K$ is a subgroup. We verify each of the group axioms:

- (G1) *Closure.* If $x, y \in H \cap K$, then $x, y \in H$ and $x, y \in K$. But then $xy \in H$ by closure of H and $xy \in K$ by closure of K . Thus $xy \in H \cap K$, and indeed the subset is closed under multiplication.
- (G2) *Associativity.* Associativity is inherited from the larger group.
- (G3) *Identity.* Since $e \in H$ and $e \in K$ (they are both subgroups), we have $e \in H \cap K$ so the identity is contained in the intersection.
- (G4) *Inverses.* If $x \in H \cap K$, then $x \in H$ and $x \in K$. But then by the axiom of inverses in each of the subgroups, we have $x^{-1} \in H$ and $x^{-1} \in K$. This implies that $x^{-1} \in H \cap K$, and we're done.

On the other hand, $H \cup K$ need not be a subgroup. Consider $D_2 = \{e, r, m, rm\}$. We have $\{e, r\}$ and $\{e, m\}$ are subgroups. But $\{e, r\} \cup \{e, m\} = \{e, r, m\}$ is not a subgroup.

5. Is it ever possible to express a group G in the form $G = H \cup K$ for two of its proper subgroups H, K ? It is ever possible to express a group G in the form $H \cup K \cup L$ for three of its proper subgroups H, K, L ?

Solution: For the first part, the answer is no. Suppose that $G = H \cup K$ for two proper subgroups H, K . Since both H, K are proper, we can choose $h \in H \setminus K$ and $k \in K \setminus H$. We now ask where the element hk must land. If $hk = h' \in H$, then we have $k = h^{-1}h' \in H$, which is a contradiction. Similarly, if $hk = k' \in K$, then we have $h = k'k^{-1} \in K$ which is a contradiction. Thus this can never happen.

On the other hand, the Klein four-group D_2 can be written as $D_2 = \{e, r\} \cup \{e, m\} \cup \{e, rm\}$ which is the union of three of its subgroups.

Cosets and Lagrange's theorem

6. Find all the subgroups of the quaternion group Q_8 . Construct the left cosets of one of the order four subgroups, and verify that they all have the same size, and partition the group.

Solution: The only possible orders of the subgroups are 1, 2, 4, 8. Going through the possibilities, we have:

- The order one subgroup is just $\{1\}$.
- Any element of an order two subgroup must be either the identity, or have order two. This implies that the only possible choice is $\{1, -1\}$.
- The elements of the order four subgroups can have at most order four. If i is in a subgroup, then this implies that $i^2 = -1, i^3 = -i$ are also in the subgroup. Similarly if $-i$ is in a subgroup, then this implies that $(-i)^2 = -1$ and $(-i)^3 = i$ are also in the subgroup. Thus the only subgroup of order four containing either i or $-i$ is $\{1, i, -1, -i\}$. Similarly we have subgroups $\{1, j, -1, -j\}$ and $\{1, k, -1, -k\}$. These are all the possibilities.
- The only subgroup of order eight is Q_8 .

Choose the subgroup $H = \{1, i, -1, -i\}$. The left cosets are:

$$H = \{1, i, -1, -i\}, \quad jH = \{j, -k, -j, k\}.$$

Certainly these have the same size, and they partition the group.

7. Consider the set of matrices of the form:

$$A = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix},$$

where a, b, c are integers taken modulo 5.

- Show that these matrices form a finite group G , and determine its order.
- Show that the subset given by setting $a = c$ defines an Abelian subgroup H . Find the order of H , and the number of distinct left cosets of H in G .
- Find the set of all elements of G whose square is the 3×3 identity matrix. Is the subset of G defined by $b \neq 0$ a subgroup of G ?

Solution:

(a) We check each of the group axioms in turn:

(G1) *Closure*. The product of two of these matrices is given by:

$$\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & a' & b' \\ 0 & 1 & c' \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & a+a' & b'+ac'+b \\ 0 & 1 & c'+c \\ 0 & 0 & 1 \end{pmatrix},$$

hence this is still a matrix of the given form.

(G2) *Associativity*. Matrix multiplication is associative.

(G3) *Identity*. The identity I is contained in this set, with $a = b = c = 0$.

(G4) *Inverses*. Looking at the multiplication formula above, we see that we have inverses:

$$\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -a & ac-b \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{pmatrix}.$$

This is still a matrix of the given form, so inverses are all contained in the set.

Therefore this set forms a group under matrix multiplication. Further, there are 5 choices for a , 5 choices for b , and 5 choices for c , giving a total of 125 group elements.

(b) For this part, take $a = c$ and $a' = c'$ in the above group multiplication law. Checking each of the group axioms in turn:

(G1) *Closure*. The group multiplication law becomes:

$$\begin{pmatrix} 1 & a & b \\ 0 & 1 & a \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & a' & b' \\ 0 & 1 & a' \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & a+a' & b'+aa'+b \\ 0 & 1 & a+a' \\ 0 & 0 & 1 \end{pmatrix},$$

so indeed the subset is closed.

(G2) *Associativity*. Matrix multiplication is associative.

(G3) *Identity*. The identity has $a = c = 0$, so is still contained in this subset.

(G4) *Inverses*. Specialising the inverse formula with $a = c$, we see that:

$$\begin{pmatrix} 1 & a & b \\ 0 & 1 & a \\ 0 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -a & a^2 - b \\ 0 & 1 & -a \\ 0 & 0 & 1 \end{pmatrix},$$

so this subset contains all of its inverses.

Thus this is a subgroup. Now we only have 5 choices for a , and 5 choices for b . This gives a total of 25 elements in H . By Lagrange's theorem, we have $125/25 = 5$ distinct left cosets of H in the group.

(c) Using the group multiplication formula, we have that:

$$\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}^2 = \begin{pmatrix} 1 & 2a & 2b + ac \\ 0 & 1 & 2c \\ 0 & 0 & 1 \end{pmatrix}.$$

For this to be the identity, we require $a = b = c = 0$, so the only relevant element is the identity itself.

The subset with $b \neq 0$ is *not* a subgroup, because it does not contain the identity.

8. Find the group table of the group generated by the matrices:

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

Find all subgroups of the group, and give a complete set of left cosets of one of the proper, non-trivial subgroups.

Solution: First, we construct the group table. We observe that:

$$A^2 = I, \quad B^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad B^3 = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}, \quad B^4 = I.$$

We also note that:

$$AB = \begin{pmatrix} i & 0 \\ 0 & i \end{pmatrix} = BA.$$

so this group is Abelian, with all elements expressible uniquely in the form $A^\alpha B^\beta$ where $\alpha = 0, 1$ and $\beta = 0, 1, 2, 3$. The group table is therefore:

$\cdot$	I	B	B^2	B^3	A	AB	AB^2	AB^3
I	I	B	B^2	B^3	A	AB	AB^2	AB^3
B	B	B^2	B^3	I	AB	AB^2	AB^3	A
B^2	B^2	B^3	I	B	AB^2	AB^3	A	AB
B^3	B^3	I	B	B^2	AB^3	A	AB	AB^2
A	A	AB	AB^2	AB^3	I	B	B^2	B^3
AB	AB	AB^2	AB^3	A	B	B^2	B^3	I
AB^2	AB^2	AB^3	A	AB	B^2	B^3	I	B
AB^3	AB^3	A	AB	AB^2	B^3	I	B	B^2

The group has size eight, so by Lagrange's theorem, the possible subgroup sizes are 1, 2, 4 and 8. Examining the table, we see that the orders of the group elements $\{I, B, B^2, B^3, A, AB, AB^2, AB^3\}$ are given by $\{1, 4, 2, 4, 2, 4, 2, 4\}$. This helps us to find all the subgroups:

- At order one, there is only $\{I\}$, and at order eight, there is only the group itself.
- At order two, there are $\{I, B^2\}$, $\{I, A\}$ and $\{I, AB^2\}$. The left cosets of $H = \{I, A\}$ are:

$$H = \{I, A\}, \quad BH = \{B, AB\}, \quad B^2H = \{B^2, AB^2\}, \quad B^3H = \{B^3, AB^3\}.$$

- At order four, we can choose to include an order four element. If the subgroup contains B , we must have B^2, B^3 too, so the subgroup must be $\{I, B, B^2, B^3\}$. If the subgroup contains B^3 , it must contain $(B^3)^2 = B^2$, and then $(B^3)^3 = B$, so is the same subgroup.

If the subgroup contains AB , we must have $(AB)^2 = B^2$ in the subgroup too, and then $(AB)^3 = AB^3$. Thus the subgroup must be $\{I, AB, B^2, AB^3\}$. Similarly if the subgroup contains AB^3 , then it must contain the inverse AB , so must be $\{I, AB, B^2, AB^3\}$.

This leaves only the possibility of a subgroup which contains the identity along with the order two elements, which would necessarily be $\{I, A, B^2, AB^2\}$. Constructing the group table for this potential subgroup, we see it is given by:

$\cdot$	I	A	B^2	AB^2
I	I	A	B^2	AB^2
A	A	I	AB^2	B^2
B^2	B^2	AB^2	I	A
AB^2	AB^2	B^2	A	I

which we observe is the group table of D_2 . Thus this is indeed a subgroup.

9. Using Lagrange's theorem, show that:

- any group of order 101 is Abelian;
- if a group contains elements with orders 6 and 10, then it must have order at least 30;
- the proper subgroups of any group of order 21 are all cyclic.

Solution:

- Let G be a group of order 101. Since 101 is prime, by Lagrange's theorem all non-identity elements have order 101. Choose x to be a non-identity element. Since it has order 101, it must generate the whole group. Hence the group is cyclic, isomorphic to C_{101} , and thus Abelian.
- The group order must contain 6 and 10 as factors, by Lagrange's theorem. The lowest common multiple of 6 and 10 is 30, so this is the smallest possible order.
- The only factors of 21 are 3, 7, so the sizes of the possible subgroups are 3, 7 respectively. But 3, 7 are both prime, implying that the subgroups (if they exist) are necessarily cyclic and isomorphic to C_3, C_7 .

10. When can a finite group G contain an element of order greater than $|G|/2$?

Solution: The order of an element divides the order of the group, so in this case, such an element g must have order $|G|$. Therefore, the group must be cyclic, generated by g .

Classification of groups

11. Show that if a non-trivial group has no proper non-trivial subgroups, then it must be cyclic and of prime order.

Solution: Since the group is non-trivial, choose x different from the identity. If x has infinite order, then it generates a subgroup:

$$\{\dots, x^{-2}, x^{-1}, e, x, x^2, \dots\}.$$

This must be all of G , since G has no proper non-trivial subgroups. But then $\{\dots, x^{-2}, e, x^2, \dots\}$ is a proper non-trivial subgroup, contradiction. Thus x cannot have infinite order.

Since x has finite order, let it be n , and consider the subgroup generated by x given by $\{e, x, x^2, \dots, x^{n-1}\}$. This subgroup must be the whole group, since the group has no proper non-trivial subgroups. Hence the group is cyclic. Further, cyclic groups have subgroups for each of the divisors of their order. Hence the cyclic group must have prime order, proving the result.

12. Classify all groups which have exactly three subgroups.

Solution: Let G be a group with three subgroups. Then the subgroups must be $\{e\}, H, G$ with H a proper non-trivial subgroup. This implies that there is some element $g \notin H$. Consider the subgroup $\{e, g, g^2, \dots\}$ generated by g . This is not the identity, and not H , so it must be all of G . Thus G must be a cyclic group.

Now, G cannot be $\mathbb{Z}$ because this has subgroups $2\mathbb{Z}, 3\mathbb{Z}, 4\mathbb{Z}$, etc, so more than three. Therefore G must be a finite group. By the classification of subgroups of cyclic groups, we know that there is a unique subgroup of C_n for each divisor of n . Thus n must have exactly three positive divisors; this implies that $n = p^2$ for some prime p . Thus the group is of the form C_{p^2} for p a prime. This works since $\{e\}, C_p, C_{p^2}$ are the only subgroups.

13. Without using Cauchy's theorem, explain why a group of even order must contain an element of order two.

Solution: We can partition the group into sets containing elements and their inverses, $\{x, x^{-1}\}$. The only elements that appear as singleton sets are the identity $\{e\}$ and any order two elements. Therefore the order of the group can be written as:

$$|G| = |\{e\}| + \sum_{x \text{ order } 2} |\{x\}| + \sum_{x \text{ not order } 2} |\{x, x^{-1}\}| = 1 + \#(\text{order } 2) + 2\#(\text{not order } 2)$$

Since $|G|$ is even, it follows that the number of order two elements must be odd; that is, there must be at least one order two element.

14. (a) Show that if a group G has two distinct subgroups H, K , both of prime order, then $H \cap K = \{e\}$.
 (b) Now suppose that the order of the group G is $2p$, where p is a prime unequal to 2.
 (i) Explain why there must be some element $x \in G$ of order 2 generating a subgroup $H = \{e, x\}$ of order 2, and some element $y \in G$ of order p generating a subgroup $K = \{e, y, \dots, y^{p-1}\}$ of order p . Explain why $H \cap K = \{e\}$.
 (ii) Explain why the left and right x -cosets of K coincide, $xK = Kx$. Deduce that $xyx^{-1} = y^k$ for some k , and hence, show that $y^{k^2} = y$. Deduce that $k = \pm 1 \pmod{p}$.
 (iii) Thus show that all groups of order $2p$ are isomorphic to either C_{2p} or D_p .

Solution:

- (a) Let H, K be distinct subgroups both of prime order, say $p, q > 1$. Observe that $H \cap K$ is a subgroup, by the result earlier on in the sheet. But then $H \cap K \leq H$ is a subgroup, so by Lagrange's theorem its order must divide p . If $|H \cap K| = p = |H|$, then $H \cap K = H$, which gives $H \subseteq K$. Then $H \leq K$, so p must divide q , giving $p = q$, contradicting the subgroups being distinct. Hence, $|H \cap K| = 1$, giving $H \cap K = \{e\}$ as required.
- (b) (i) In the group of order $2p$, Cauchy's theorem guarantees the existence of elements x, y of order 2 and p respectively. These give the required subgroups. Since $2, p$ are distinct primes, we must have that $H \cap K = \{e\}$ by part (a).
 (ii) The left cosets of K are K, xK , which partition the group since $x \notin K$. Similarly the right cosets K, Kx have this property. But then since they partition the group, we must have $xK = Kx$ as required. It follows that $xy = y^k x$ for some k , giving $xyx^{-1} = y^k$ as required. We can therefore deduce that $y^{k^2} = (xyx^{-1})^k = xy^k x^{-1} = x(xy x^{-1})x^{-1} = y$, since x has order two. Multiplying by y^{-1} on both sides, we have $y^{k^2-1} = e$, which implies that $k^2 - 1$ is a multiple of the order p . Since p divides $k^2 - 1 = (k - 1)(k + 1)$, we must have that either p divides $k - 1$ or $k + 1$, since p is prime. It follows that $k = \pm 1 \pmod{p}$.
 (iii) In the case $k = 1$ modulo p , we have $xyx^{-1} = y$, i.e. $xy = yx$. In particular, this implies that $(xy)^{2p} = x^{2p}y^{2p} = e$, so that xy has order at most $2p$. Further $(xy)^k = x^k y^k = e$ if and only if $y^k = x^{-k}$, which occurs if and only if $y^k = e$ and $x^k = e$ individually. Thus this is the first time this occurs, since the lowest common multiple of $2, p$ is $2p$. Hence the group is generated by xy , and is isomorphic to the cyclic group C_{2p} .
 On the other hand, if $k = -1$ modulo p , the group is generated by x, y subject to $x^2 = e$, $y^p = e$ and $xy = y^{-1}x$. These are precisely the defining relations of the dihedral group D_p , hence we're done.

15. If every proper subgroup of a group is cyclic, must the group be cyclic?

Solution: No - an easy counterexample is D_3 , whose only proper subgroups are $\{e\}$, $\{e, r, r^2\}$, $\{e, m\}$, $\{e, rm\}$, $\{e, r^m\}$, which are all cyclic. Similarly Q_8 works!

Bonus questions

16. (*) Is it true that every infinite group has an infinite proper subgroup? Provide a proof or counterexample as appropriate.

Solution: A counterexample is the set of rotations of the circle by $2\pi n/2^k$, where n is an integer, and $k \geq 0$ is a non-negative integer. Certainly this is closed, since:

$$\frac{2\pi n}{2^k} + \frac{2\pi m}{2^l} = \frac{2\pi(n \cdot 2^l + m \cdot 2^k)}{2^{k+l}},$$

so two successive rotations result in the same form of rotation. It is associative, because it involves composition of symmetries. The identity is a rotation by zero angle ($n = 0$), and the inverse of $2\pi n/2^k$ is $-2\pi n/2^k$.

Now let H be a subgroup of this group. Note that if H contains the rotation $2\pi n/2^k$ with n odd, then we also include the rotation $2\pi/2^k$ in the group if we can perform the rotation $2\pi n/2^k$ a total of r times to give:

$$\frac{2\pi rn}{2^k} = \frac{2\pi}{2^k} + 2\pi s$$

for some s . This occurs if and only if $rn = 1$ modulo 2^k ; this is guaranteed, since n and 2^k are coprime. Thus if H contains $2\pi n/2^k$ for n odd, then it must contain the rotation $2\pi/2^k$.

If H contains the rotation $2\pi n/2^k$ with n even, we can always reduce to some $2\pi n'/2^{k-1}$ with $n = 2n'$, and continue in this fashion until we obtain an element of the form $2\pi n/2^k$ with n odd. Thus we may assume that if H contains any non-zero rotations, then it contains rotations of the form:

$$\frac{2\pi}{2^k}$$

for some k . We now split into two cases:

- If H contains a *largest* such k , suppose that $2\pi n/2^l$ is any other rotation contained in H . Since $l < k$, we have:

$$\frac{2\pi}{2^l} = \frac{2\pi \cdot (2^{k-l}n)}{2^k},$$

so we see that this rotation can be generated from $2\pi/2^k$. Thus H is isomorphic to the cyclic group generated by $2\pi/2^k$, and thus must be finite.

- Else, H does not contain a largest k , and we can rotate by arbitrarily small angles. We claim that this gives all other rotations. Given any rotation by $2\pi/2^k$, there must exist some $l \geq k$ such that $2\pi/2^l$ is contained in K . But then:

$$\frac{2\pi}{2^k} = \frac{2\pi 2^{l-k}}{2^l},$$

so $2\pi/2^k$ is contained in H . Thus all rotations of the form $2\pi/2^k$ are contained in H , and this obviously generates all of the original group.